SYNERGETICS
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Java 11 New Features

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Agenda

- Local Variable Type Inference (LVTI)
- Private Method in Interface
- Modules
- Annotations
- Secure Coding
- Localisation

Local Variable Type Inference

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- Java is a strongly typed language i.e., you must specify the type of a variable when introducing it.
- However, since Java 10, we have the option, in certain scenarios, of using the keyword var instead of the type. In these situations, the compiler infers the type e.g., var x = “ abc ”; // x is a String
- Remember that the variables must be local and must be initialized on the line where they are declared.

Local Variable Type Inference

- LVTI.java Example

Private method in interface

- Introduced in Java 9, interfaces can now have private methods.
- These private methods can be both static and non static.
- As they are private , they are accessible within the interface only.
- As with classes, you cannot access a non static interface method from a static method.

Private method in interface

- Benefits are:
  - reduce code duplication
  - improve code encapsulation

Private method in interface

- PrivateMethodInterface.java Example

Modules

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- Modules, introduced in Java 9, provide an extra layer of encapsulation by enabling us to group related packages.
- From a high level, modules enable us to specify well defined boundaries/dependencies in our code base.
- As a developer, you can specify which packages are accessible outside the module and also the module dependencies

Why Modules?

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- Improved access control:
  - In addition to private, package, protected and public now we have the ability to restrict packages to certain other packages.

- Improved large scale structure of applications:
  - boundaries/dependencies in your code base.
  - hide implementation details.
  - improved decoupling.

Modules

- By default, all classes in the packages in a module are strongly encapsulated i.e. no other module can use these classes, even if they are public classes.
- Package names and module names live in separate namespaces i.e. you can have the same identifiers without any conflict (name clash).
- A module contains one or more packages plus a module info.java file.

Services

- Modules are tightly coupled due to the use of requires , exports and the knowledge and creation of specific types.
- Services enable the decoupling of modules.
- Rather than modules directly coupling to other modules, there is now a middle layer, the service registry which abstracts the consumers from the producers.
- The service (interface) contract , which is hosted by the registry, is what binds the consumers and producers together.

Annotations

- Metadata is information about information.
- Annotations, via metadata, enable us to add value to our code.
- We can annotate (assign metadata) to classes, methods variables etc..
- Though optional, when used they must be used correctly.

Annotations

- Annotations operate much like interfaces. In fact, their origins are in interfaces. Their “elements” look like abstract methods.
- Annotation names are case sensitive; however, it is common practice to start the name with an uppercase letter.
- Like interfaces, annotations can be applied to unrelated classes.

Annotations

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- As stated earlier, annotations have a lot in common with interfaces; for example, a marker annotation has no elements (a marker interface has no methods). Annotations, as with interfaces, can be applied to unrelated classes.

- In fact, we annotate our annotation with @interface

- public @interface MyAnnotation {} // marker annotation

Annotations

- An annotation element is an attribute of the annotation. The elements (attributes) can have values.

```
public @interface MyAnnotation{
        int myElement (); // looks like an abstract
}
```

- Remember that annotations have their origins in interfaces. Essentially, the JVM translates the above element into an interface method and the annotation itself as an implementation of the interface.

Annotations

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- There are annotations that can be applied to annotations
- As opposed to applying built in annotations to types (methods, interfaces, fields etc..); we can actually apply built in annotations to annotations themselves.
- For example, you might want to limit where your annotation can be used or you may want your annotation information included in Javadoc documentation.

Annotations

- @Target limits the types that the annotation can be applied to.
- The types are specified as an array of ElementType enum values.

Annotations

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ElementType value	Scope (what it applies to)
TYPE	Interfaces, enums, classes, annotations.
METHOD	Method declarations
PARAMETER	Constructor and method parameters
FIELD	Instance and static variables
CONSTRUCTOR	Constructor declarations
LOCAL_VARIABLE	Local variables
ANNOTATION_TYPE	Annotations
TYPE_USE	Anywhere there is a Java data type. This includes where types are *used* e.g. object creation with *new*.

Annotations

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- In generics, we encountered “type safety” where certain information is discarded by the compiler when converting source code into .class file.
- Similarly, annotations may be discarded at compile time, at runtime or not at all. We can control when this happens by using the @Retention annotation.
- We specify the level of retention using an enum value from RetentionPolicy

Annotations

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RetentionPolicy value	Description
SOURCE	Source file only, compiler discards it.
CLASS	Stored in the .class file but not available at runtime. This is the default compiler behaviour.
RUNTIME	Stored in the .class and available at runtime (via reflection).

```
import java.lang.annotation.Retention;
import java.lang.annotation.RetentionPolicy;

@Retention(RetentionPolicy.SOURCE)  // annotation discarded by the compiler
@interface Mouse{}                  // i.e. not in .class file

@Retention(RetentionPolicy.RUNTIME) // annotation stored in .class file and
@interface Keyboard{}               // available at runtime (via reflection)
```

Data Integrity

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Accessibility

- limit access as much as possible --“principle of least privilege”.
- instance variables/methods should be private

Restrict extensibility

- prevent subclassing by marking the class as final

Data Integrity

- Immutable objects are objects that cannot be changed after Creation.
- They are secure objects and use the following guidelines:
  - Do not provide any “setter” methods.
  - Make all the fields private and final
  - Prevent subclassing (prevents overriding):
    - make the class final
    - make the constructor private and provide a public static factory method e.g. “createNewInstance”
  - Prevent subclassing (prevents overriding):
    - immutable types e.g. String , ok
    - mutable types e.g. StringBuilder , do NOT share references
    - i.e . use “defensive copying” and “advanced encapsulation”

Common Injection

- SQL Injection
  - where user input retrieves unexpected results.
  - protection provided by PreparedStatement with bind variables (and not Statement).

Localisation

- Localisation involves ensuring that your program is adaptable to the location from which it is being executed.
- This involves translating strings, use of different spellings and formatting of dates/numbers for that locale.
- A locale, in its simplest terms, is a language/country pairing.

Localisation

- To create/select a locale that is not the default locale, there are 3 popular options:

- constructors
  - pass in a language only or both a language and a country
- built in constants
  - the Locale class provides constants for the most popular locales
- builder design pattern
  - flexible specify the properties you want, in any order
  - locale is built at the end

Resource Bundles

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- Resource bundles contain locale specific data.
- They take the format of a map with keys and values. The values are locale specific.
- Resource bundles can be created in 2 ways:
  - java files extend ListResourceBundle
  - property files text files with key value pairs

What is Hashing ?

It is the process of converting an object into an integer value. The integer value helps in indexing and faster searches.

What is HashMap?

HashMap is a part of the Java collection framework. It uses a technique called Hashing. It implements the map interface. It stores the data in the pair of Key and Value. HashMap contains an array of the nodes, and the node is represented as a class. It uses an array and LinkedList data structure internally for storing Key and Value. There are four fields in HashMap.

What is HashMap?

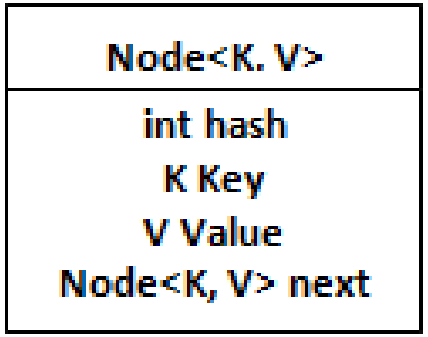

Figure: Representation of a Node

1. **hashCode():** This is the method of the object class. It returns the memory reference of the object in integer form. The value received from the method is used as the bucket number. The bucket number is the address of the element inside the map. Hash code of null Key is 0.
2. **Buckets:** Array of the node is called buckets. Each node has a data structure like a LinkedList. More than one node can share the same bucket. It may be different in capacity.

Allocation of Nodes in Bucket

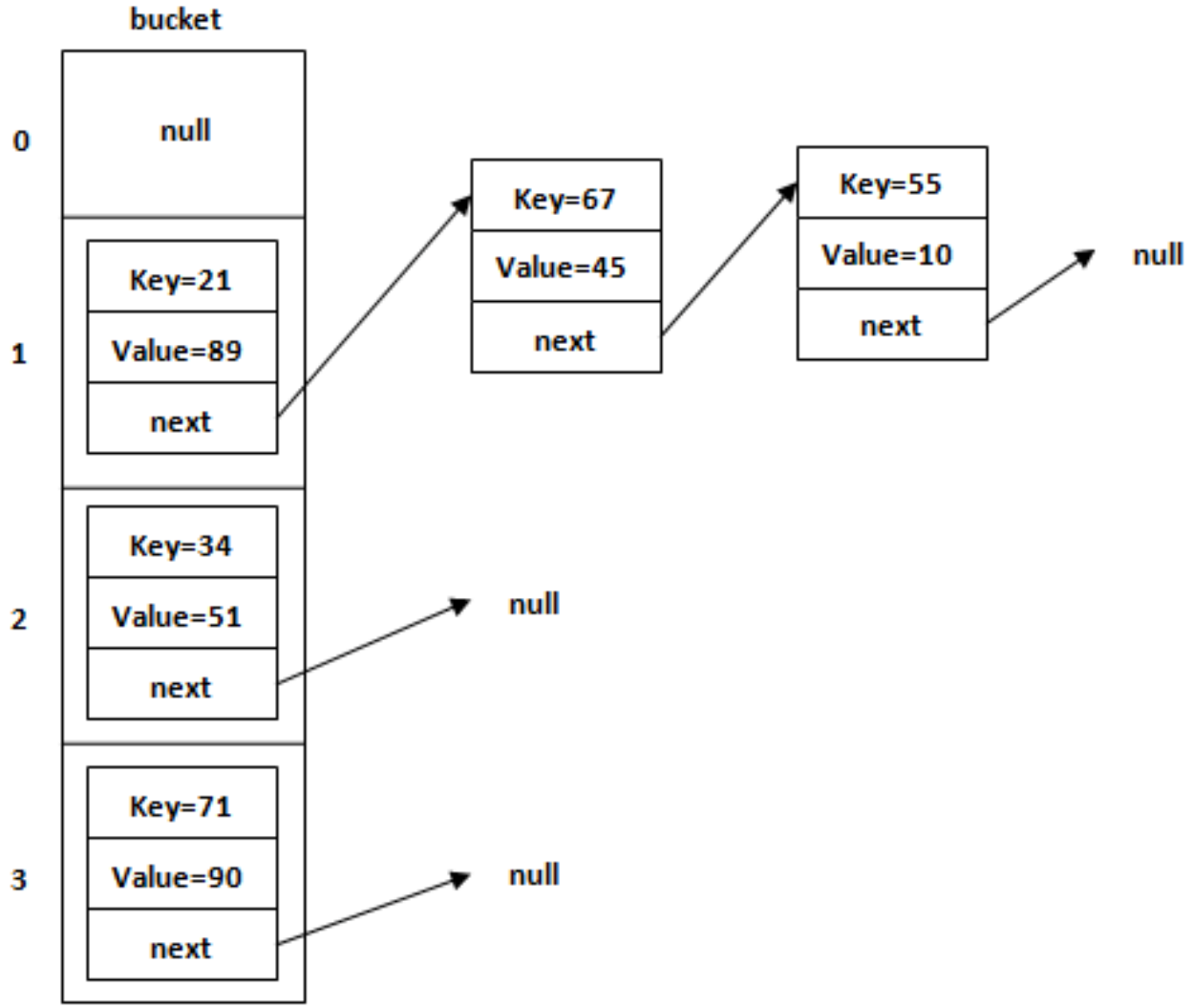

Figure: Allocation of nodes in Bucket

Insert (Key, Value) in HashMap

- To insert the Key and Value pair in the HashMap put() method is used. The default size of HashMap is 16 (0 to 15).

```
Map <String ,Integer> map = new HashMap<>();

map.put ("aaa",1);
map.put("abc",2);
```

- At which index the Key, value pair will be saved into HashMap. When we call the put() method, then it calculates the hash code of the Key **"aaa".**
- Suppose the hash code of "aaa" is **2657860**.
- To store the Key in memory, we have to calculate the index.

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Calculating Index

- If hashcode for each object is generated then it run out the memory so modular operations are performed internally and index is calculated.
- Index minimizes the size of the array. The Formula for calculating the index is:

Index = hashcode(Key) & (n-1)

- Where n is the size of the array. Hence the index value for "aaa" is:

Index = 2657860 & (16-1) = 4

2657860
aaa
1
null

0
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15

Hash Collution

- This is the case when the calculated index value is the same for two or more Keys. Let's calculate the hash code for another Key **“abc"** Suppose the hash code for “abc" is **same as 2657860**. To store the Key in the memory, we have to calculate index by using the index formula.
- The index is **4 after computed**.
- In this case, equals() method check that both Keys are equal or not. If Keys are same, replace the value with the current value.
- Otherwise, connect this node object to the existing node object through the LinkedList. Hence both Keys will be stored at index 4.

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Thank You

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